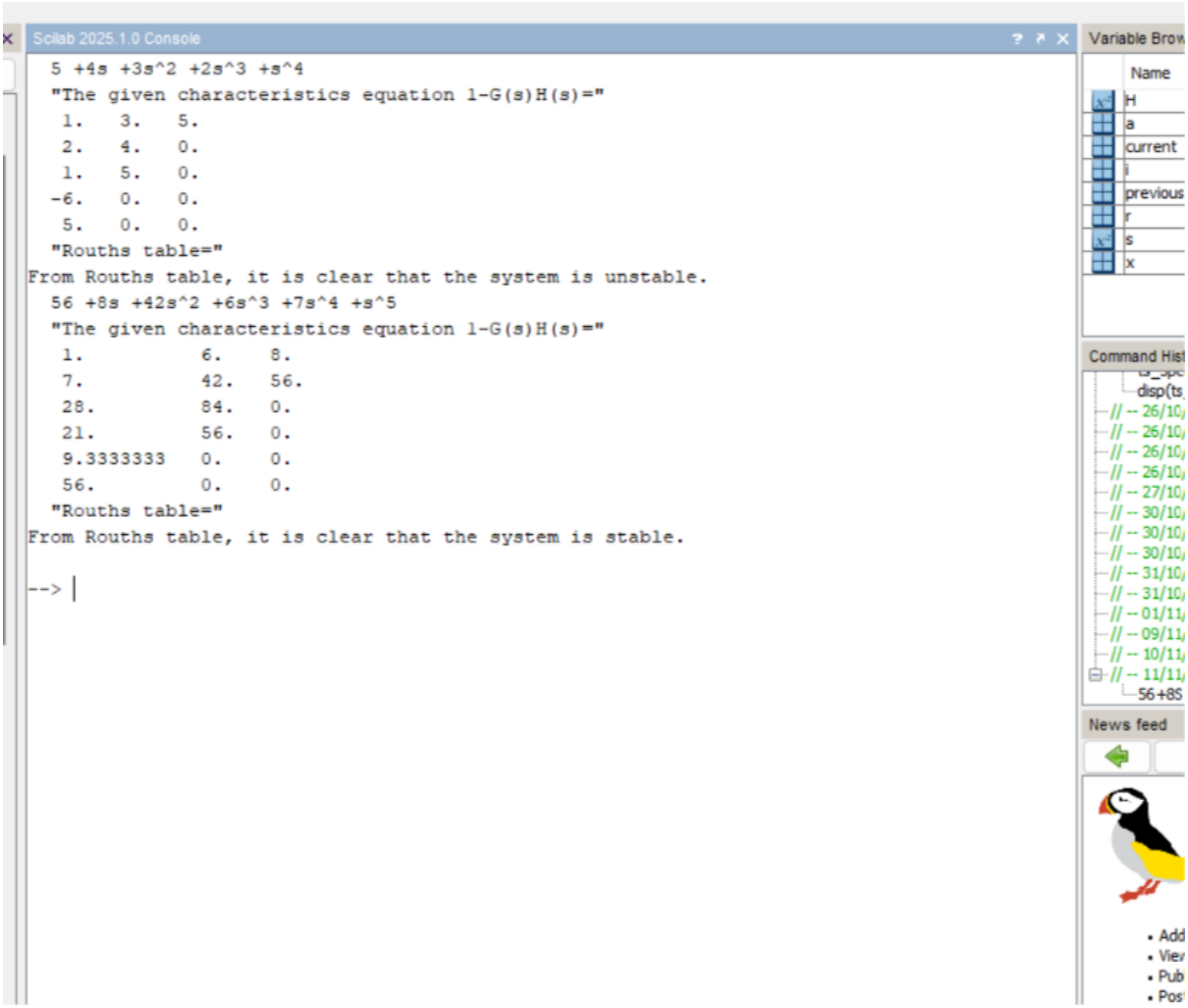
PROGRAM CODE :

```
clear;
clc;
xdel(winsid());
mode(0);
s=%s;
H=s^4+2*s^3+3*s^2+4*s+5;
//H=s^5+7*s^4+6*s^3+42*s^2+8*s+56;
disp(H,'The given characteristics equation 1-G(s)H(s)=');
c=coeff(H);
len=length(c);
r=routh_t(H);
disp(r,"Routhstable=");
x=0;
for i=1:len
if(r(i,1)<0)
x=x+1;
end
end
if(x>=1)
printf("From Rouths table, it is clear that the system is unstable.")
else
printf("From Rouths table, it is clear that the system is stable.")
end
```

OUTPUT:


Scilab 2025.1.0 Console
5 +4s +3s^2 +2s^3 +s^4
"The given characteristics equation 1-G(s)H(s)="
1. 3. 5.
2. 4. 0.
1. 5. 0.
-6. 0. 0.
5. 0. 0.
"Rouths table="
From Rouths table, it is clear that the system is unstable.
56 +8s +42s^2 +6s^3 +7s^4 +s^5
"The given characteristics equation 1-G(s)H(s)="
1. 6. 8.
7. 42. 56.
28. 84. 0.
21. 56. 0.
9.3333333 0. 0.
56. 0. 0.
"Rouths table="
From Rouths table, it is clear that the system is stable.
-->
Variable Brow
Name
H
a
current
i
previous
r
s
x
Command Hist
disp(ts
// -- 26/10/
// -- 26/10/
// -- 26/10/
// -- 26/10/
// -- 27/10/
// -- 30/10/
// -- 30/10/
// -- 30/10/
// -- 31/10/
// -- 31/10/
// -- 01/11/
// -- 09/11/
// -- 10/11/
// -- 11/11/
56+8S
News feed